

STABIO, Switzerland

WMO: 067710

Lat: 45.8435N

Lon: 8.9324E

Elev: 352

StdP: 97.16

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-5.8	-17.0	0.9	3.0	-14.2	1.1	4.0	5.3	8.7	4.4	8.0	1.0	230	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	30.9	21.4	29.5	20.8	28.2	20.1	22.7	28.8	21.9	27.6	21.1	26.4	1.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	16.1	25.3	20.0	15.3	24.5	19.2	14.6	23.7	68.6	28.9	65.5	27.7	62.7	26.5	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.2	3.4	2.7	DB	-9.9	32.9	2.5	1.4	-11.7	34.0	-13.1	34.8	-14.5	35.6	-16.3	36.7	(4)
(5)			WB	-10.6	24.3	2.4	1.1	-12.3	25.1	-13.7	25.7	-15.1	26.4	-16.8	27.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	2.3	3.5	7.7	11.3	15.7	19.9	21.9	21.3	17.0	12.6	7.1	2.9		
	DBStd	7.52	2.86	3.05	3.25	3.04	2.75	2.93	2.40	2.41	2.75	3.05	3.26	3.04		
	HDD10.0	854	241	181	84	21	1	0	0	0	0	12	94	221		
	HDD18.3	2619	498	414	331	211	89	17	3	4	57	179	336	479		
	CDD10.0	1572	1	0	12	60	177	297	368	349	209	91	8	1		
	CDD18.3	295	0	0	0	0	8	63	112	95	16	1	0	0		
	CDH23.3	2509	0	0	3	11	102	587	967	721	112	7	0	0		
	CDH26.7	656	0	0	0	1	11	156	282	194	11	1	0	0		
(14) Wind	WSAvg	0.9	0.8	0.9	1.1	1.1	1.0	1.0	1.0	0.9	0.8	0.8	0.8	0.8		
(15) Precipitation	PrecAvg	1589	75	72	79	139	180	152	145	146	171	163	194	75		
	PrecMax	2380	230	285	225	319	498	417	306	315	448	433	680	230		
	PrecMin	972	3	3	5	13	44	17	45	16	15	10	0	0		
	PrecStd	369	64	67	57	83	99	84	74	76	110	102	164	62		
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.5	17.2	22.8	24.6	28.1	31.9	32.7	32.7	28.0	23.7	17.7	14.2		
		MCWB	8.2	8.2	11.2	15.0	18.2	21.6	22.2	22.4	19.7	15.9	11.2	6.0		
	2%	DB	11.5	13.9	19.4	22.4	26.0	30.1	31.0	30.4	26.3	21.3	15.4	11.3		
		MCWB	5.5	6.9	10.3	13.6	17.1	20.9	21.4	21.7	19.1	15.6	10.6	6.0		
	5%	DB	9.5	11.9	17.3	20.6	24.2	28.6	29.5	28.8	24.5	19.5	13.6	9.7		
		MCWB	4.5	5.9	9.6	12.6	16.2	20.1	20.8	20.7	17.7	14.6	10.1	5.7		
	10%	DB	7.7	9.8	15.1	18.6	22.4	26.8	28.2	27.3	22.8	17.7	12.3	8.1		
		MCWB	3.9	4.6	8.5	11.5	15.3	19.2	20.2	20.0	16.8	14.1	9.5	5.3		
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	9.3	12.5	15.9	19.6	23.4	23.8	23.6	21.2	17.9	13.5	9.4		
		MCDB	13.2	14.9	19.9	23.1	25.9	29.6	29.7	29.7	26.1	22.0	15.6	10.6		
	2%	WB	7.0	7.9	11.3	14.5	18.2	22.0	22.7	22.5	20.0	16.9	12.2	8.0		
		MCDB	9.8	12.1	17.4	20.8	24.2	27.8	28.9	28.6	24.6	19.6	13.8	9.5		
	5%	WB	5.8	6.8	10.4	13.4	17.2	21.0	21.9	21.7	19.0	16.0	11.3	6.9		
		MCDB	8.1	10.2	15.3	18.9	22.7	26.9	27.6	27.3	22.8	18.3	12.7	8.4		
	10%	WB	4.7	5.7	9.5	12.4	16.2	20.1	21.1	21.0	18.1	14.9	10.3	5.8		
		MCDB	6.7	8.4	13.7	17.1	21.0	25.5	26.4	25.9	21.3	16.9	11.6	7.5		
(36) Mean Daily Temperature Range	5% DB	MDBR	9.5	10.8	12.1	11.4	10.9	11.3	11.6	11.0	10.3	9.0	8.6	9.1		
		MCDBR	12.8	15.1	16.9	15.4	14.3	13.6	13.2	13.1	12.6	11.8	11.1	11.9		
	5% WB	MCWBR	8.1	9.3	9.3	7.8	6.8	5.9	5.5	5.8	6.1	6.7	7.0	7.7		
		MCWBR	10.7	13.1	14.6	13.2	12.5	12.2	12.1	11.9	10.6	9.5	8.0	8.7		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.358	0.410	0.460	0.464	0.483	0.464	0.457	0.445	0.422	0.362	0.320		
	taud		2.339	2.262	2.160	2.062	2.105	2.100	2.161	2.189	2.184	2.198	2.325	2.381		
	Ebn at Noon		764	803	806	798	809	794	804	793	763	715	706	728		
	Edh at Noon		77	101	129	155	154	156	145	136	125	107	79	67		
(44) All-Sky Solar Radiation	RadAvg		1.50	2.36	3.65	4.49	5.29	6.14	6.22	5.30	4.02	2.46	1.48	1.25		
	RadStd		0.20	0.32	0.49	0.58	0.46	0.42	0.43	0.37	0.31	0.35	0.25	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.42	N/A	N/A	+0.82	+0.43	N/A	N/A	N/A	+65	+59
(47)	Regional (102 neighbors)	N/A	N/A	N/A	+0.72	+0.40	+0.32	N/A	N/A	+60	+16

Nomenclature: See separate page